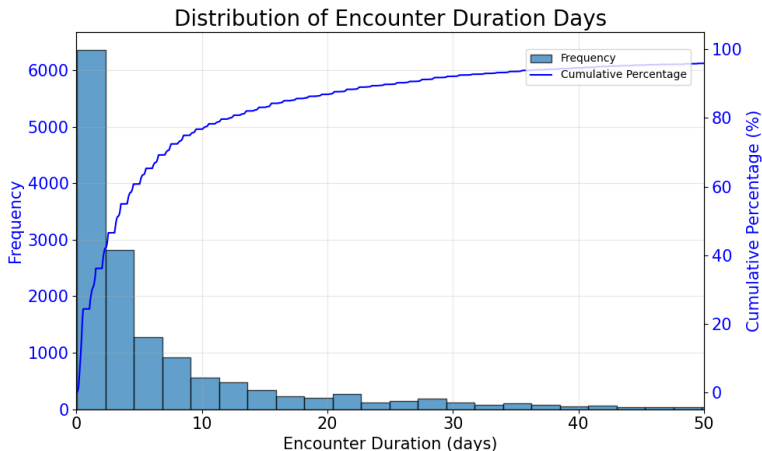


Results from length-of-stay and procedure prediction models

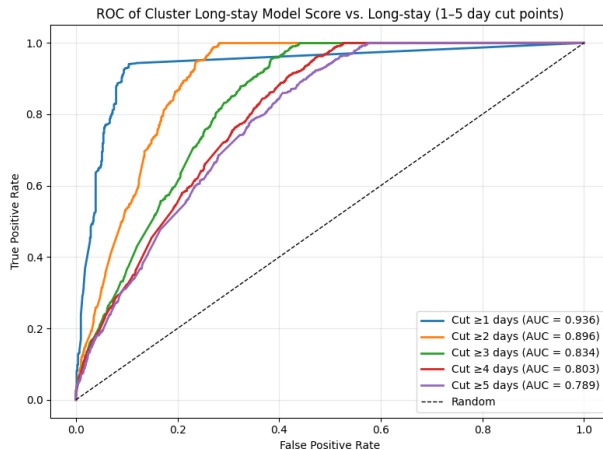
May 2, 2026

Encounter duration distribution



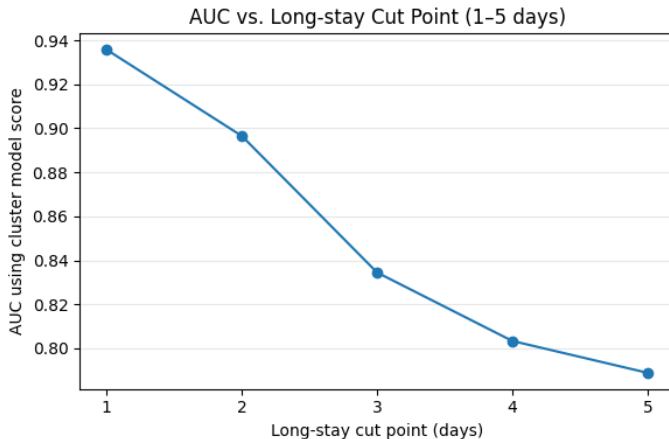
Shows the empirical distribution of encounter duration in days, with a cumulative percentage curve to motivate the downstream long-stay and patient-load prediction task. Source: `cleaned_models.ipynb`, cell 2.

Long-stay model ROC with additional features



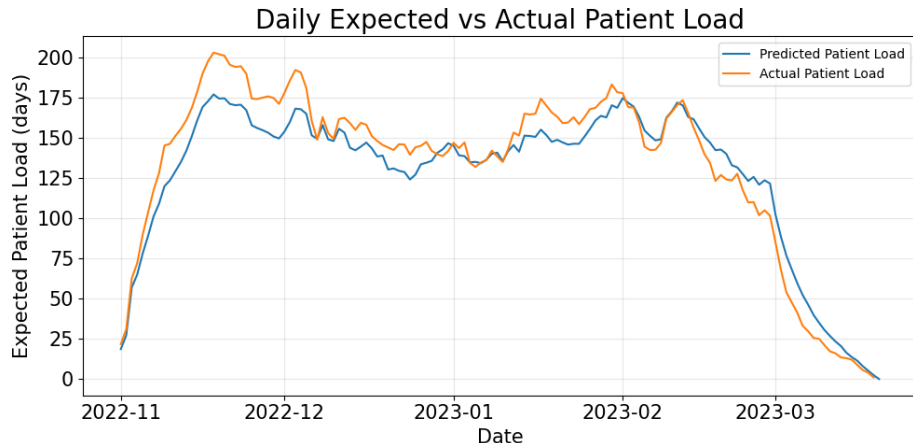
Summarizes discrimination for the long-stay classifiers. Source logic: `cleaned_models.ipynb`, cells 3–7.

Long-stay model cut points with additional features



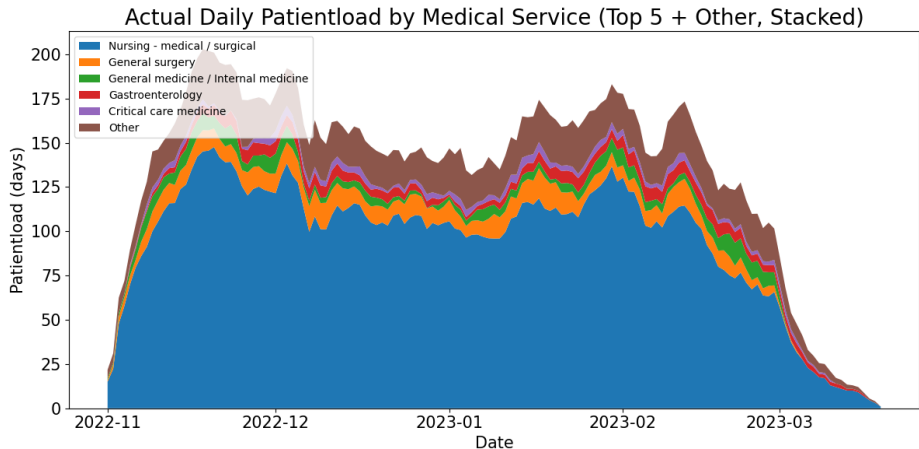
Shows operating cut points for translating long-stay probabilities into day-level workload estimates. Source logic: `cleaned_models.ipynb`, cells 5-8.

Daily expected versus actual patient load



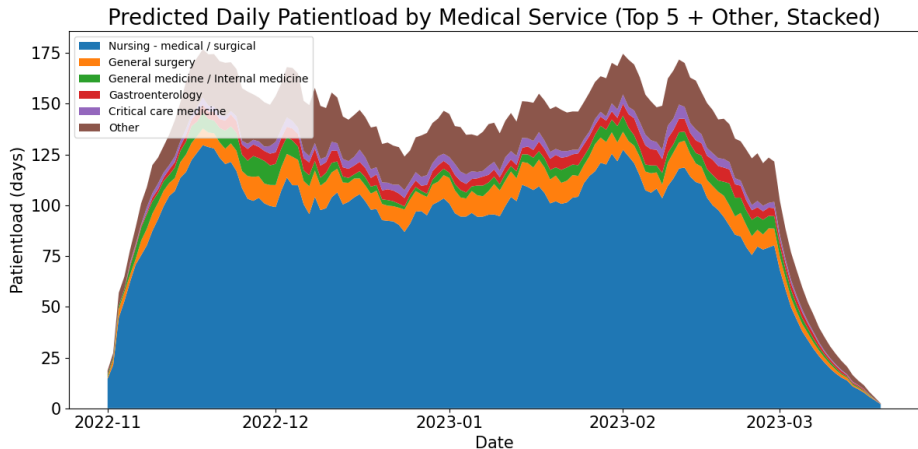
Compares expected daily patient load from summed long-stay probabilities against actual workloads reconstructed from observed encounter durations. Source: `cleaned_models.ipynb`, cells 8, 11–12.

Actual patient load by medical service



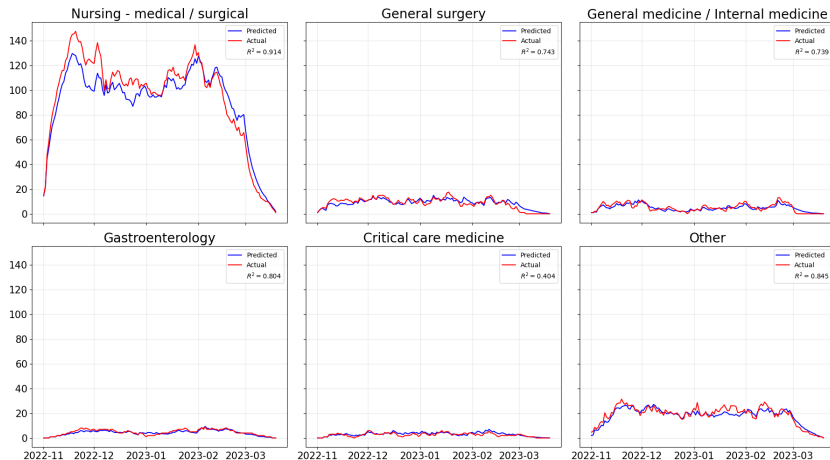
Breaks actual daily workload into the top five medical services plus an “Other” category, using observed encounter duration contributions. Source: `cleaned_models.ipynb`, cell 9.

Predicted patient load by medical service



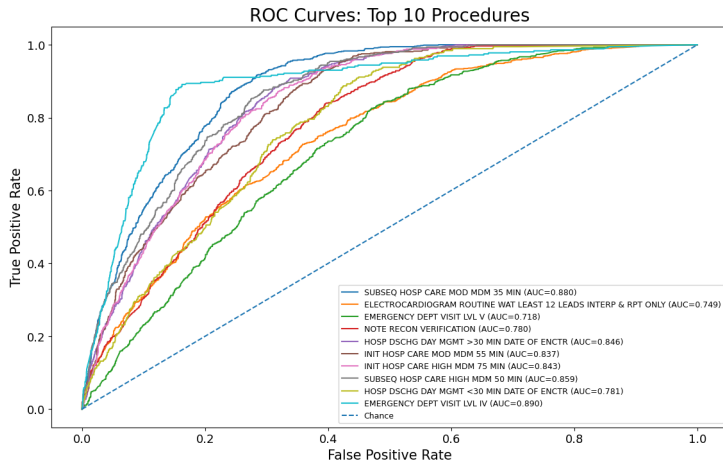
Shows predicted daily workload by medical service by summing each encounter's probability of extending into future days.
Source: `cleaned_models.ipynb`, cell 9.

Predicted versus actual patient load by service



Compares predicted and actual service-level patient load over time, with R^2 reported per service panel. Source: `cleaned_models.ipynb`, cell 10.

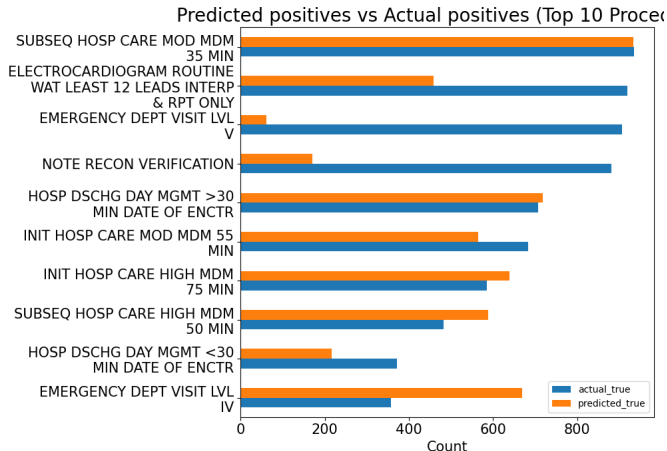
ROC curves for top 10 procedures



Evaluates one classifier per common procedure using ROC curves and AUC values for the top ten procedure labels.

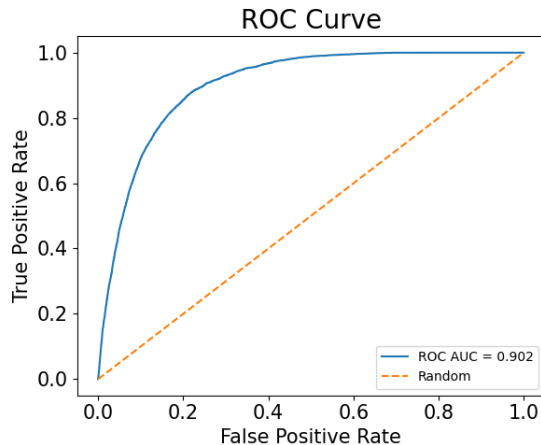
Source: `procedures.ipynb`, cells 6–8.

Predicted versus actual positives for top procedures



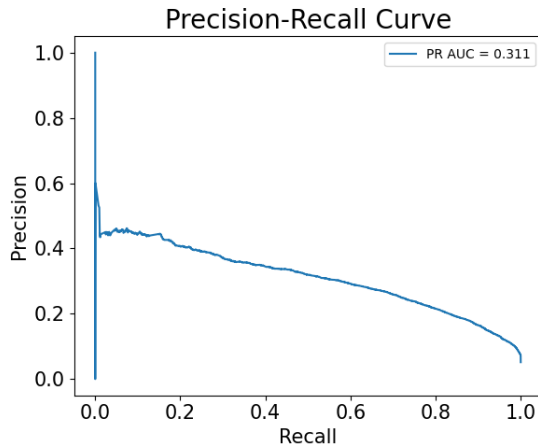
Compares the number of actual positive procedure labels to thresholded model predictions for the top ten procedures.
Source: `procedures.ipynb`, cell 10.

Day-level procedure model ROC



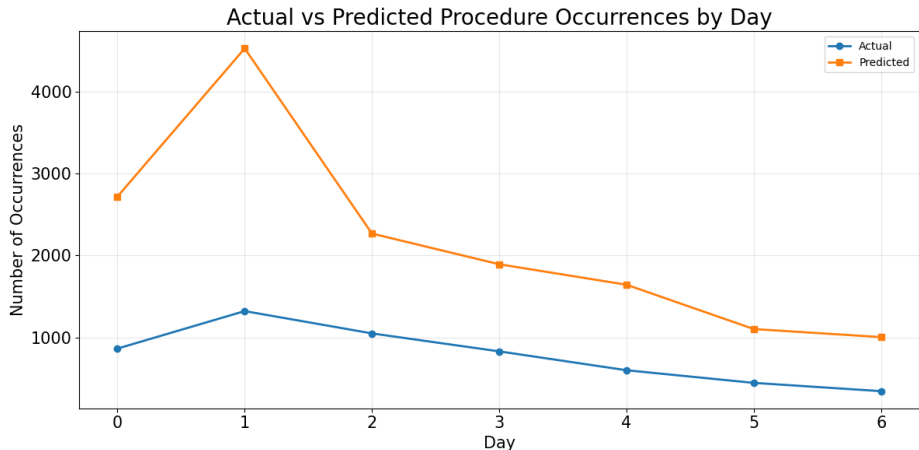
Evaluates the day-level procedure-occurrence model, which predicts whether selected procedures occur on a given day of an encounter. Source: `procedures.ipynb`, cells 12–13.

Day-level procedure model precision-recall



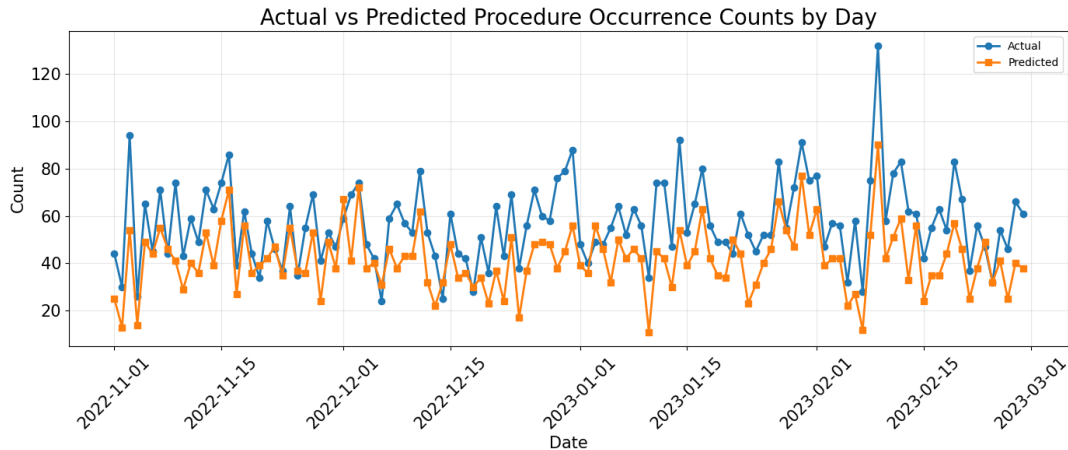
Evaluates precision and recall across thresholds for the day-level procedure-occurrence model. Because positive procedure days are rare, PR AUC should be interpreted relative to the positive-rate baseline. Source: `procedures.ipynb`, cells 12–13.

Procedure occurrences by encounter day



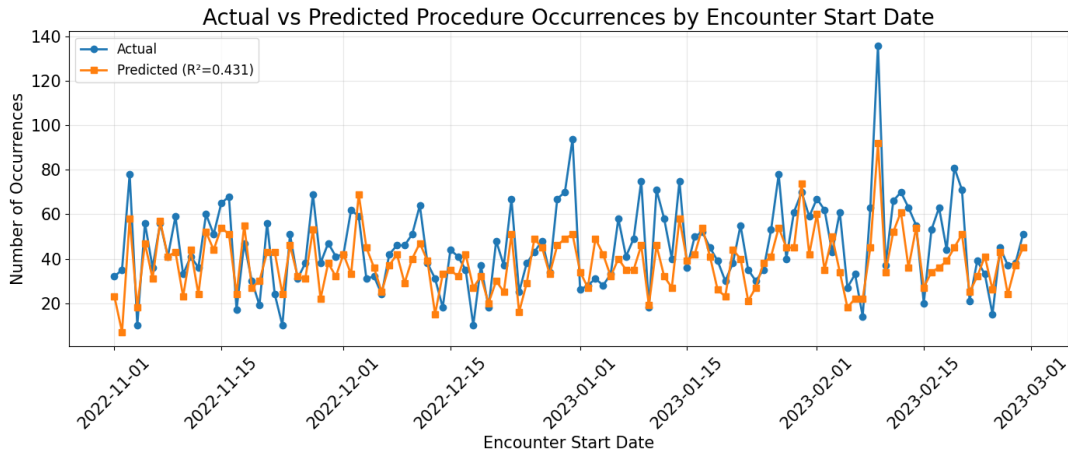
Aggregates actual and predicted procedure occurrences by encounter day to show whether the model captures within-stay procedure timing. Source: `procedures.ipynb`, cell 14.

Procedure occurrence counts by calendar day



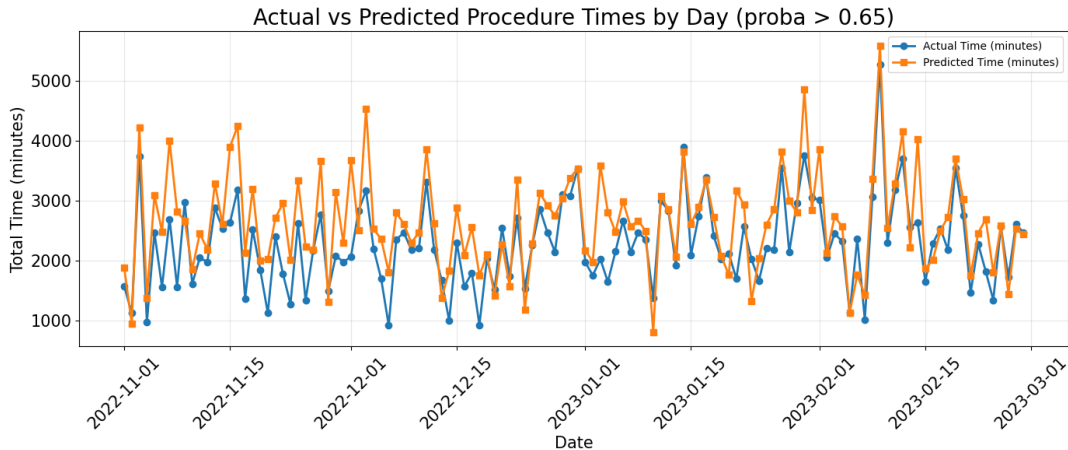
Compares daily actual versus predicted counts of top-ten procedure occurrences using encounter start dates. Source: `procedures.ipynb`, cell 16.

Procedure occurrences by encounter start date



Shows actual and predicted procedure occurrences over encounter start dates, including an R^2 summary for temporal fit.
Source: `procedures.ipynb`, cell 17.

Estimated procedure time by day



Converts predicted and actual procedure labels into estimated minutes using a hand-coded procedure-time dictionary, then aggregates by day. Source: `procedures.ipynb`, cells 24–26.